

BASIC RESEARCH – TECHNOLOGY

Laser-activated Versus Sonic-activated Irrigation for Removing Residual Filling Materials in Mandibular Molars: A Micro-Computed Tomography Study

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ABSTRACT

Introduction: Advanced irrigation plays a crucial role in nonsurgical retreatment cases, as these procedures require the removal of existing obturation materials while preserving dentin. The aim of this study was to compare the effectiveness of laser-activated versus sonic-activated irrigation in removing residual obturation material in the mesial canals of mandibular molars with isthmuses during nonsurgical root canal retreatment.

Methods: Forty mesial canals of extracted mandibular molars were instrumented and obturated with gutta-percha and a bioceramic sealer. Retreatment was performed using rotary instruments, followed by a final irrigation protocol consisting of 2.5% NaOCl, 17% EDTA, and a final rinse of 2.5% NaOCl using either sonic-activated or laser-activated irrigation (LAI). Micro-computed tomography scans were obtained before and after final irrigation. The volume of obturation material was measured to calculate the median percentage of remaining material in each group. **Results:** The LAI group showed a significantly lower median percentage of remaining filling material than the sonic group (73.9% vs. 93.8%, $P = .0165$). However, neither irrigation protocol completely eliminates residual obturation material from the canals. **Conclusion:** LAI was more effective at removing remaining obturation material compared to sonic-activated irrigation. (*J Endod* 2026; ■:1–7.)

KEY WORDS

Laser-activated irrigation; retreatment; sonic-activated irrigation

Nonsurgical retreatment is a fundamental procedure in endodontics, aiming to remove the source of persistent infection and facilitate the healing of post-treatment diseases. The emergence of post-treatment diseases is predominantly attributed to residual bacteria within the root canal system, often obscured due to inaccessible areas or overlooked anatomical complexities during the initial treatment^{1,2}. Prior to achieving thorough disinfection and obturation of the root canal system in retreatment, it is imperative to completely remove the previously placed obturation material³. This step allows unhindered access for instruments and irrigants to reach the persistent microorganisms.

Despite significant advancements in endodontic technology, the complete removal of residual gutta-percha from root canal walls remains a challenge especially in well-obturated canals. Traditionally, gutta-percha is removed mechanically with stainless steel hand files and nickel-titanium rotary instruments aided by solvents, heat, and ultrasonics³. Complete removal of residual gutta-percha and sealer in areas such as the apical third and the isthmuses has proven to be difficult. Previous studies have shown none of the mechanical or adjunctive techniques such as hand and rotary instrumentation, XP-Endo Finisher, and addition of solvent could remove gutta-percha entirely from the root canal system⁴⁻⁶. Additionally, as the use of bioceramic sealers become more widespread, its retreatability remains a valid concern. Previous research has shown difficulties in regaining working length when the

SIGNIFICANCE

Although the Er,Cr:YSGG LAI protocol was more effective than the sonic-activated irrigation at removing residual root filling material, neither of the methods consistently achieved complete removal of gutta-percha and sealer during endodontic retreatment.

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apical 2 mm of the canal was filled by bioceramic sealer only. A recent study showed that bioceramic sealer can penetrate into isthmuses especially after activation with advanced irrigation, potentially making it more challenging to remove⁷.

The erbium, chromium: yttrium-scandium-gallium-garnet (Er,Cr:YSGG) laser has been used as an adjunct in root canal irrigation. It has been reported that Er,Cr:YSGG laser activation was significantly more effective in removing debris compared to sonic activation in minimally instrumented root canal systems⁸. The working mechanism of a Er,Cr:YSGG laser in root canal irrigation is attributed to cavitation effects, which induces high speed fluid motion into and out of the canal. Specifically, low energy pulsed erbium produces optical cavitation in the irrigant to form vapor bubbles in microseconds. As the bubbles form and collapse at high speed, fluid motion is created to disrupt biofilms on the root canal walls as well as deep in the dentinal tubules. Studies showed this enhancement in irrigant penetration into dentinal tubules improved canal disinfection^{9,10}. To the best of our knowledge, there is no previous research on evaluating the efficacy of Er,Cr:YSGG laser-activated irrigation (LAI) as an adjunct tool in removing residual gutta-percha and bioceramic sealer. Therefore, the objective of this study was to compare the efficacy of LAI and sonic-activated irrigation in removing residual gutta-percha and bioceramic sealer in the mesial canals of mandibular molars in nonsurgical root canal retreatment.

MATERIALS AND METHODS

Ethics

The study protocol was submitted to the Texas A&M College of Dentistry Institutional Review Board (protocol: 152866) and was granted "exempt" status.

Sample Size Calculation

A priori sample size calculation was performed using G*Power software (Version 3.1.9.6, Heinrich Heine University, Düsseldorf, Germany) for Mac. The effect size (Cohen's $d = 1.61$) was derived from a previous study¹¹, which compared the percentage of remaining gutta-percha among different rotary systems. Using a two-tailed t-test with a significance level of 0.05 and power of 0.95, the minimum required sample size was calculated to be 12 specimens per group (total $n = 24$). Two specimens in the laser group developed cracks due to desiccation during micro-computed tomography (micro-

CT) imaging and were therefore excluded from the final analysis. This resulted in 19 specimens in the laser group and 21 samples in the sonic group.

Sample Selection

Freshly extracted mandibular first and second molars were collected and stored in phosphate-buffered solution. All teeth were radiographically evaluated and any teeth with open apex, internal root or external root resorption, or caries extended into the root were excluded. When present, coronal caries were removed and restored with composite.

Only teeth with mesial roots of Weine type II and III configuration were included in the study¹². Preoperative micro-CT (μ CT 35; Scanco Medical, Bassersdorf, Switzerland) were obtained to confirm the canal configuration, the presence of an isthmus, the absence of calcification, and an apical diameter no greater than 0.20 mm. The root curvature of each specimen was measured using Mimics Innovation Suite (v26; Materialise, Brussels, Belgium), and mesial roots with curvatures between 10° and 30° were included.

Initial Root Canal Preparation, Irrigation, and Obturation

All treatments were performed by a second-year endodontic resident under an operating microscope (Zeiss OPMI Pico, Jena, Germany) using 0.6 magnification changer factor and following the manufacturer's instructions. The resident had completed a comprehensive 1-day laser training course and subsequently used the laser system in clinical cases prior to this study.

Working length (WL) was established by confirming patency with a manual size 10 K-file to the apical foramen, then subtracting 1 mm from this measurement. The canals were instrumented using DCTaper rotary files (SS White, Lakewood, NJ) up to a master apical file size of 20/V06. 2.5% NaOCl was used for irrigation using side-vented 30-gauge NaviTip needle (Ultradent, South Jordan, UT).

All canals were obturated to the WL with laser-verified Fine-Fine fitted gutta-percha cones (SS White, Lakewood, NJ) and EndoSequence BC Sealer HiFlow (Brasseler, Savannah, GA) using a single-cone technique¹³. After obturation was completed, the pulp chamber was cleaned with a cotton pellet soaked in 70% alcohol, and the pulp chamber was temporized with Teflon tape and OpalDam Green (UltraDent, South Jordan, UT). All teeth were placed at 37°C

under 100% humidity for 72 hours to allow the sealer to set.

Root Canal Retreatment

A Vortex Blue (Dentsply Sirona, Charlotte, NC) 25 of 04 rotary file was used to instrument the canals to WL. Small amounts of chloroform were used to soften the gutta-percha throughout instrumentation. A size 10 K-file was used to confirm patency. Canals were irrigated with 5 mL of 2.5% NaOCl throughout the instrumentation procedure. A radiograph was taken to verify the bulk of the gutta-percha has been removed. After instrumentation was completed, the pulp chamber was cleaned with a cotton pellet soaked in 70% alcohol, and the pulp chamber was temporized with Teflon tape and OpalDam Green. A micro-CT scan was taken to record the baseline residual gutta-percha and sealer before using adjunct irrigation techniques.

Adjunctive Irrigation Protocols

Specimens were randomly (<http://www.random.org>) assigned into either sonic-activated or laser-activated irrigation groups.

In the SI group, the EndoActivator (Dentsply Sirona, Charlotte, NC) was operated at 10,000 cycles per minute, and a 15/02 tip was used in an up and down motion to within 1 mm of the WL. Irrigation and activation were performed in three cycles of 30-second duration per canal: the first cycle with 1 mL of 2.5% NaOCl, the second cycle with 1 mL of 17% EDTA, and the third cycle with 1 mL of 2.5% NaOCl. The total activation of irrigants for each canal was 90 seconds.

In the LAI group, WaterLase iPlus (Biolase, Lake Forest, CA) was used. This Er,Cr:YSGG laser operates at a wavelength of 2780 nm. The radial firing tips RFT2-17 mm (0.25 mm diameter) were used with the following settings: 1.25 W power, 25 mJ energy, air: 0, water: 0, 50 Hz pulse, Mode: H. The tip was placed to the middle third of the canal. The laser was activated while withdrawing the tip at a rate of 2 mm/s. The radial disinfection step was repeated multiple times until each irrigation was activated for 30 seconds per cycle, per canal. The first cycle was with 1 mL of 2.5% NaOCl, the second cycle was with 1 mL 17% EDTA, and the third cycle was with 1 mL of 2.5% NaOCl. The total activation of irrigants for each canal was 90 seconds.

The canals were dried using paper points. The chamber of the teeth was temporized with Teflon tape and OpalDam Green before a micro-CT scan to record the

residual obturation materials after using adjunct irrigation techniques.

Micro-CT Acquisition and Reconstruction Procedures

Specimens were scanned using a micro-CT system (μ CT 35; Scanco Medical, Bassersdorf, Switzerland) at 15 μ m nominal voxel size. The scanner setting was set at 55 kVp energy and 145 μ A intensity with an integration time of 200 ms.

The acquired projection images were reconstructed, and the preoperative 3D models of the samples were rendered by the Scanco software (Scanco Medical). The initial values for volume (mm^3) and surface area (mm^2) were calculated using the Scanco software. The volume and surface area of interest were selected extending from the furcation level to the apex of the root.

Micro-CT Analysis

Mimics Innovation Suite (v26; Materialise, Brussels, Belgium) was used for segmentation and volume analysis. Masks were generated using thresholding and region-growing operations. When necessary, the "Edit Masks" function was used to manually refine the boundaries of the obturation material. The segmented obturation material in each root was then converted into a 3D model. Volumes of mesial canals before and after adjunctive irrigation were measured, and the percentage of residual obturation material was calculated using the following formula: (Postirrigation obturation volume/Preirrigation obturation volume) $\times$ 100.

Statistical Analysis

Normality of the continuous data (isthmus width and root curvature) was assessed using the Shapiro–Wilk test. As both datasets followed a normal distribution ($P > .05$), group comparisons were performed using independent t-tests. The distribution of Weine canal configurations (categorical data) was compared using the chi-square test. For the percentage of remaining gutta-percha, the data did not follow a normal distribution (Shapiro–Wilk $P < .05$); therefore, a

nonparametric Mann–Whitney U test was used to compare the laser and sonic groups.

RESULTS

There were no statistically significant differences between the laser and sonic groups in any of the evaluated root canal morphology parameters, including isthmus width, root curvature, and Weine canal type (Table 1). This indicates that the groups were comparable at baseline.

The median percentage of remaining filling material in the laser group was 73.9% (IQR: 31.7%), whereas the sonic group exhibited a higher median of 93.8% (IQR: 18.7%) (Fig. 1). This difference was statistically significant ($P = .0165$), indicating superior efficacy of laser irrigation in removing gutta-percha from the canal system (Table 2).

Representative micro-CT images of residual obturation material before and after adjunctive irrigation are shown for both the laser group and the sonic group (Fig. 2).

DISCUSSION

The primary goal of nonsurgical endodontic retreatment is to promote periapical healing by complete removal of the previous root filling materials, thorough disinfection, and re-obturation of the root canal system. The removal of previous obturation materials especially bioceramic sealer has been shown to be challenging^{14,15}. None of the traditional techniques was able to completely remove all obturation materials^{3,16}. Additionally, a major limitation of mechanical debridement and traditional irrigation is limited contact and penetration of irrigant to the entire canal system. As a result, attempts have been made to utilize irrigation activation techniques to increase irrigant distribution in the root canal system and improve efficacy of disinfection^{17,18}. Laser-assisted irrigation has shown its efficacy on the removal of bacterial biofilms, smear layer, and dentinal debris in previous *in vitro* studies^{8,10,19}.

This study examined the effects of Er:Cr:YSGG laser on retreatment of extracted mandibular molars obturated with gutta-

percha and bioceramic sealer in an *in vitro* setting. The results of this study indicated superior efficacy of laser irrigation in removing gutta-percha from the canal system compared to sonic activation. The difference in irrigation effectiveness can be attributed to the activation mechanisms employed by these adjunctive irrigation protocols operate through fundamentally different principles. The EndoActivator delivers up to 167 Hz low frequency sonic energy to the irrigant via mechanical agitation²⁰. A study demonstrated sonic activation performs as well as or better than passive ultrasonic irrigation at eliminating microorganisms from curved and straight root canals²¹. However, this energy transmission becomes compromised when the polymer tip contacts canal curvature or constriction²². In contrast, the Er:Cr:YSGG laser transmits low-level light energy through hydrodynamic mechanisms. Even when the laser tip is positioned at the orifice level, the fluid streaming and pressure waves remain effective to remove substantial amounts of debris beyond the curved canal and enhance irrigant penetration to dentin²².

The results of this study are comparable to previous studies that investigated the efficacy of LAI on retreatment of teeth obturated with gutta-percha and bioceramic sealer. These studies have shown additional laser application led to a significant reduction in the amount of the remaining filling materials after rotary instrumentation²³. Photon-induced photoacoustic streaming based Er:YAG laser was effective in reducing residual filling materials compared to conventional irrigation methods, particularly in the apical third of root canals where access is most challenging²⁴. When it is compared with ultrasonic-activated irrigation, a study found that neither technique was able to eliminate the residual materials, although laser activation significantly improved removal compared to ultrasonic and needle irrigation²⁵. While other studies found that laser combined with NaOCl was significantly more effective at removing filling remnants than sonic and ultrasonic techniques²⁶. On the contrary, some other studies showed that photon-induced photoacoustic streaming based LAI did not significantly improve the removal of filling material compared to conventional needle irrigation, passive ultrasonic irrigation, and XP-endo Finisher²⁷. Additionally, a study reported passive ultrasonic irrigation was more effective than Er:YAG laser activation for removing MTA Fillapex from root canal walls²⁸.

This is the first study to evaluate the use of an Er:Cr:YSGG laser in removing residual obturation materials during endodontic

TABLE 1 - Comparative Analysis of Root Canal Morphology Parameters between Sonic and Laser Groups

Parameters	Sonic group (n = 21)	Laser group (n = 19)	P value
Isthmus width (mm)	Mean: 2.95 Std Dev: 0.85	Mean: 3.39 Std Dev: 0.66	.0773 (t-test)
Root curvature (deg)	Mean: 24.14 Std Dev: 5.24	Mean: 22.79 Std Dev: 5.99	.4785 (t-test)
Weine type	Weine Type 2: 15 Weine Type 3: 6	Weine Type 2: 9 Weine Type 3: 10	.2195 (Chi Square)

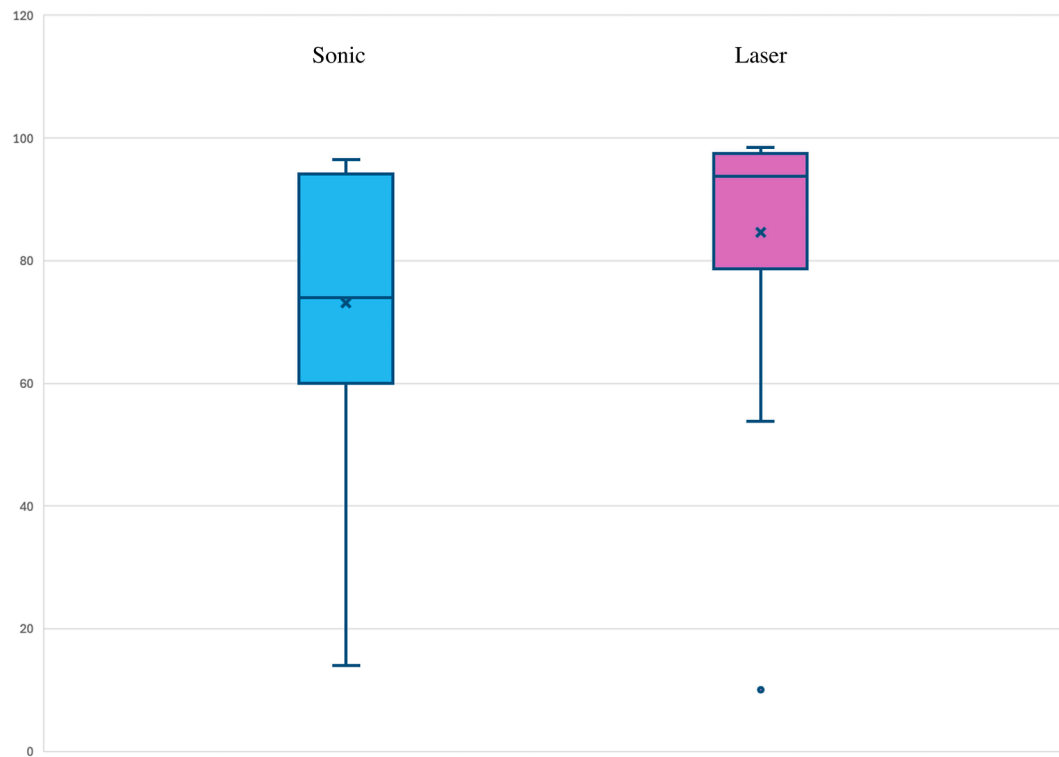


FIGURE 1 – Box plot showing the percentage of residual gutta-percha after adjunctive irrigation protocol. Blue, sonic-activated irrigation; Purple, laser-activated irrigation.

retreatment. Previous studies in this area have primarily used different wavelength Er:YAG or Nd:YAG lasers. Compared to Er:YAG and Nd:YAG lasers, Er,Cr:YSGG laser produces higher surface temperatures and has better absorption of water and hydroxyapatite, which potentially provide more efficient tissue removal and more bacterial reduction²⁹. This study also differs in that the laser tip was placed directly into the middle third of the canal, rather than being limited to the pulp chamber as in most prior may theoretically enhance hydrodynamic effects by directing energy closer to the remaining obturation material. It is also the first study to specifically evaluate removal of residual materials from mesial canals of mandibular molars. These areas are especially difficult to clean, as gutta-percha and bioceramic sealers tend to accumulate and harden in isthmuses and are often resistant to removal by mechanical instrumentation⁷.

In this study, sonic activation was chosen as an experimental group instead of ultrasonic because ultrasonic irrigation tip is

limited straight portion of the canal and cannot be placed to an ideal length in a minimally prepared curved canal³⁰. Additionally, cutting of dentin during active ultrasonic irrigation can lead to ledging of the canal and instrument fracture³¹. The canals in this study were initially prepared to 20/V06 and retreated to 25/04. It is important to note that the initial 20/V06 instrument had a variable taper design, which produces a smaller overall prepared canal volume than a conventional 25/04 file, while the 25/04 also provides a larger apical diameter. This allowed a conservative preparation while still providing measurable enlargement for retreatment. Although retreatment protocols in clinical practice often enlarge canals beyond 2 instrument sizes^{32,33}, this file was intentionally selected to be only slightly larger in size to ensure that enough residual gutta-percha remained in the canal for meaningful comparison between irrigation protocols³⁴. We recognize that the absence of a control group with larger preparation sizes that more closely reflect common clinical retreatment protocols can be

interpreted as a limitation. Future studies could investigate larger preparation sizes, particularly if imaging advances enable reliable differentiation between residual gutta-percha and sealer. This would provide additional insight into how canal enlargement influences activation efficacy under different retreatment conditions.

Working with extracted human teeth introduces unavoidable anatomical variability, which is recognized as a limitation of *in vitro* endodontic research. At the same time, this variability adds value because it more closely reflects the clinical complexity encountered during retreatment than standardized resin or plastic models. To reduce the influence of anatomical differences, strict inclusion criteria were applied and specimens were randomly allocated to the 2 irrigation groups. Baseline micro-CT measurements confirmed that there were no statistically significant differences in key anatomical parameters including Weine configuration, isthmus width, and root curvature, indicating that both groups were well balanced (Table 1). Although some level

TABLE 2 - Comparison of Remaining Gutta-Percha (%) between Laser and Sonic Irrigation Groups

	Median (%)	IQR (%)	Minimum (%)	Maximum (%)	P value
Sonic (n = 21)	93.8	31.7	10.0	98.4	.0165 (Mann–Whitney U test)
Laser (n = 19)	73.9	18.7	13.9	96.4	

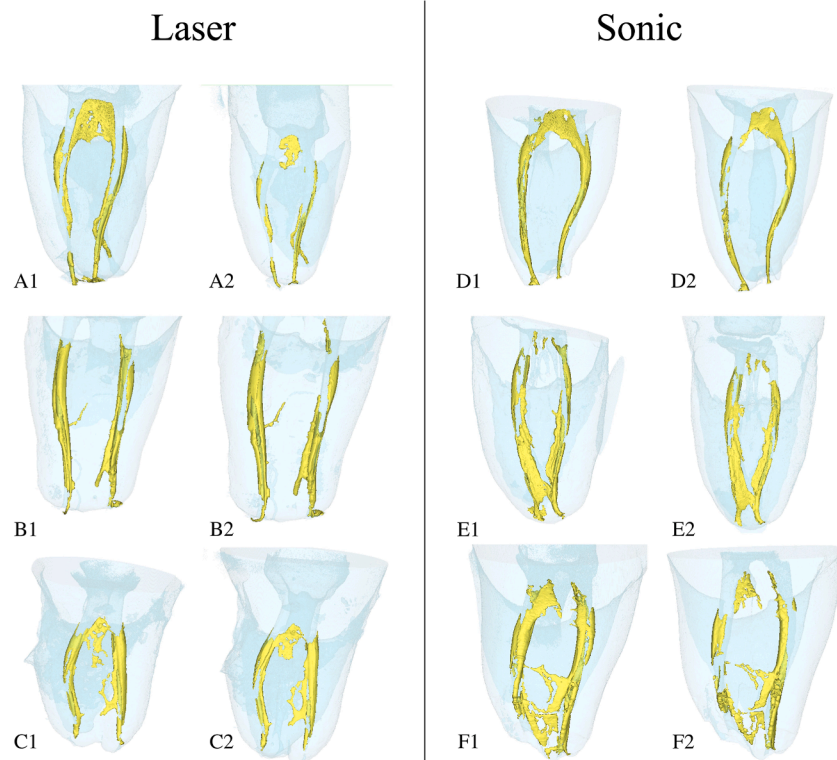


FIGURE 2 – Representative 3D reconstructed images showing residual obturation material before and after adjunctive irrigation. On the left, panels *A1*, *B1*, and *C1* illustrate preirrigation status, and *A2*, *B2*, and *C2* show postirrigation status following laser activation. On the right, panels *D1*, *E1*, and *F1* represent preirrigation status, and *D2*, *E2*, and *F2* show postirrigation status following sonic activation.

of natural variability remains, this approach supports both clinical relevance and comparability between the intervention groups. A further limitation of this study is the 72-hour sealer setting period, which was selected to match reported maximum setting times while minimizing specimen fragility;

however, future studies using longer setting intervals may help clarify how extended aging influences retreatability outcomes.

Although LAI can enhance the removal of debris and smear layer, it showed limited effect in dislodging gutta-percha in the present study. This observation may be

attributed to several factors gutta-percha is a pliable, rubber-like material that tends to absorb energy rather than fracture under stress, which may reduce the efficacy of cavitation forces. It also adheres tightly to canal walls and fills canal irregularities, making it less susceptible to detachment through fluid dynamics alone. Furthermore, cavitation effects are diminished in the constricted anatomy of root canals, particularly in the apical third, where energy transmission is less effective³⁴. Taken together, these factors may explain why cavitation-based methods, including laser activation, appear insufficient for gutta-percha removal and highlight the continued importance of mechanical instrumentation and chemical softening in retreatment procedures.

CONCLUSION

In this study, adjunctive irrigation method did not eliminate residual filling material from root canal systems. While the LAI demonstrated superior median percentage removal of residual filling material compared to the sonic-activated irrigation method with statistical significance, this improvement may not be clinically significant. The effectiveness of the LAI in removing bacteria and tissue remnants when residual filling material persists following endodontic retreatment remains uncertain and requires further investigation.

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The authors deny any conflicts of interest related to this study.

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